

[← Go Back](#)

Library

Recents viewed

[Arduino_MKRENV](#)[Arduino_MKRGPS](#)[Arduino_APDS9960](#)[Arduino_MKRIoTCarrier](#)[Home](#) / [Programming](#) / [Library](#) / [Arduino_MKRENV](#)

ON THIS PAGE

[Usage/Examples](#)[Compatibility](#)[Releases](#)[Methods](#)

+

SENSORS

Arduino_MKRENV

[ARDUINO](#) [LICENSE](#) LGPL-2.1 [V1.2.1](#) [Arduino](#) [11/08/2021](#)[Arduino](#) <info@arduino.cc><http://github.com/arduino-lib...> info@arduino.cc

Allows you to read the temperature, humidity, pressure, light and UV sensors of your MKR ENV Shield.

GO TO REPOSITORY

[Usage/Examples](#)[Compatibility](#)[Releases](#)

The Arduino MKR ENV library allows you to read the sensors on the [MKR ENV Shield](#). It manages the different interfaces used by the sensors on the shield to give you an uniform and simple set of functions to read them. The library takes care of the calculations needed to produce values in the requested units. The values returned are signed floats.

To use this library:

```
1 #include <Arduino_MKRENV.h>
```

The Arduino MKR ENV library takes care of the sensor initialization and sets its values as follows:

Absolute pressure range: 260 to 1260 hPa.

Humidity range: 0 to 100% relative humidity (rH).

Humidity accuracy: $\pm 3.5\%$ rH, 20 to +80% rH.

Temperature range -40 to 120 °C.

Temperature accuracy: ± 0.5 °C from 15 to 40 °C.

Lux range: 10 to 100,000 lux.

UVA/UVB resolution: 16-bit; unit $\mu\text{W}/\text{cm}^2$.

UVIndex: 1 to 11+.

Methods

[begin\(\)](#)[Help](#)

Initialize the sensors on the shield.

Syntax

```
1 ENV.begin()
```

Parameters

None.

Returns

1 on success, 0 on failure.

Example

```
1 if (!ENV.begin()) {  
2     Serial.println("Failed to i  
3     while (1);  
4 }
```

See also

- `end()`
- `readTemperature()`
- `readHumidity()`
- `readPressure()`
- `readIlluminance()`
- `readUVA()`
- `readUVB()`
- `readUVIndex()`

▼ `end()`

De-initialize the sensors on the shield.

Syntax

```
1 ENV.end()
```

Parameters

Returns

None.

Example

```
1 ENV.end();
```

See also

`begin()`
`readTemperature()`
`readHumidity()`
`readPressure()`
`readIlluminance()`
`readUVA()`
`readUVB()`
`readUVIndex()`

▼ `readTemperature()`

Read the temperature sensor's value. If no unit is specified as parameter, the value will be expressed in Celsius.

Syntax

```
1 ENV.readTemperature(unit)
```

Parameters

unit: FAHRENHEIT to get the temperature in Fahrenheit and CELSIUS to get the temperature in Celsius (default).

Returns

The sensor's temperature value as float in the specified unit.

Example

```
1 Serial.print("Temperature = ");
2 Serial.print(ENV.readTemperature());
3 Serial.println(" °C");
```

See also

`begin()`
`end()`
`readHumidity()`
`readPressure()`
`readIlluminance()`
`readUVA()`
`readUVB()`
`readUVIndex()`

▼ `readHumidity()`

Read the HTS221 sensor's relative humidity (rH), and returns it as a percentage. Read more about relative humidity in [this article](#).

Syntax

```
1 ENV.readHumidity()
```

Parameters

None.

Returns

Returns the relative humidity (rH) in percentage.

Example

```
1 Serial.print("Humidity    = ");
2 Serial.print(ENV.readHumidity());
3 Serial.println(" %");
```

See also

`begin()`

```
readPressure()  
readIlluminance()  
readUVA()  
readUVB()  
readUVIndex()
```

▼ readPressure()

Read the pressure sensor's value. If no unit is specified, the value will be expressed in kilopascal.

Syntax

```
1 ENV.readPressure(unit)
```

Parameters

unit: PSI to get the pressure in pound per square, MILLIBAR to get the pressure in millibar and KILOPASCAL to get the pressure in kilopascal (default).

Returns

The sensor's pressure value as float in the specified unit.

Example

```
1 Serial.print("Pressure  = ");  
2 Serial.print(ENV.readPressure());  
3 Serial.println(" kPa");
```

See also

```
begin()  
end()  
readTemperature()  
readHumidity()  
readIlluminance()  
readUVA()  
readUVB()  
readUVIndex()
```

▼ readIlluminance()

Read the light sensor's value.

Syntax

```
1 ENV.readIlluminance(unit)
```

Parameters

unit: FOOTCANDLE to get the light value in footcandle, METERCANDLE to get the light in metercandle and LUX to get the light value in lux (default).

Returns

The light sensor's value as float in the specified unit.

Example

```
1 Serial.print("Lux = ");  
2 Serial.println(ENV.readIllumina
```

See also

begin()
end()
readTemperature()
readHumidity()
readPressure()
readUVA()
readUVB()
readUVIndex()

▼ readUVA()

Read the UV sensor's UV A value.

Syntax

```
1 ENV.readUVA()
```

Parameters

None.

Returns

The UV sensor's UV A value.

Example

```
1 Serial.print("UVA = ");  
2 Serial.println(ENV.readUVA());
```

See also

`begin()`
`end()`
`readTemperature()`
`readHumidity()`
`readPressure()`
`readIlluminance()`
`readUVB()`
`readUVIndex()`

▼ `readUVB()`

Read the UV sensor's UV B value.

Syntax

```
1 ENV.readUVB()
```

Parameters

None.

Returns

The UV sensor's UV B value.

Example

```
1 Serial.print("UVB = ");
2 Serial.println(ENV.readUVB());
```

See also

`begin()`
`end()`
`readTemperature()`
`readHumidity()`
`readPressure()`
`readIlluminance()`
`readUVA()`
`readUVIndex()`

▼ `readUVIndex()`

Read the UV sensor's UV index value.

Syntax

```
1 ENV.readUVIndex()
```

Parameters

None.

Returns

The UV sensor's UV index value.

Example

```
1 Serial.print("UV Index = ");
2 Serial.println(ENV.readUVIndex());
```

See also

`begin()`
`end()`
`readTemperature()`
`readHumidity()`
`readPressure()`

`readUVB()`▼ `readLux()`

Description

Read the light sensor's value.

Syntax

```
1 ENV.readLux()
```

Parameters

None

Returns

The light sensor's value in Lux.

Example

```
1 Serial.print("Lux: ");  
2 Serial.println(ENV.readLux());
```

See also

`begin()`
`end()`
`readTemperature()`
`readHumidity()`
`readPressure()`
`readIlluminance()`
`readUVA()`
`readUVB()`
`readLux()`

Was this article helpful?



- [Project Hub](#)
[GitHub Repository](#)
[Forum](#)
- [Product Compliance](#)
[Help Center](#)
[Trademarks & Licensing](#)